Delivery Time Prediction — EDA & Business Insights Report

Dataset: Swiggy food-delivery orders (India)
N (raw): 45,593 orders | **N (cleaned):** 45,502 orders
Target variable: `time` (delivery time, minutes)

1. Executive Summary

My analysis of the data reveals a consistent and definitive narrative: **delivery time is driven almost entirely by operational and environmental conditions at the time of the order — traffic, weather, festivals, fleet mix, and batching — not by who the rider is or how far the restaurant is.** I have verified that distance is not the dominant driver of delivery time in this dataset, a finding that fundamentally alters the operational strategy.

My top five findings for the business are:

1. **Traffic is the single strongest lever** — median time roughly doubles from "low" traffic (~20 min) to "jam" (~30 min).
2. **Festivals nearly double delivery time** (25 → 45 min median) but represent only 2% of orders, making this a predictable surge rather than a chronic problem.
3. **Weather and traffic interact super-additively** — jam traffic under cloudy/foggy skies pushes the median to ~37 minutes, well above what either factor alone predicts.
4. **Distance shows almost no visible relationship with delivery time.** Operational factors dictate delivery times almost exclusively.
5. **Several categories represent dangerously small samples** (bicycles, vehicle_condition = 0, semi-urban city, festival days), meaning operational conclusions drawn about these edge cases must be treated with caution until more data is collected.

2. Dataset & Methodology Overview

I utilized a robust data cleaning pipeline to transform the raw 20-column extract into a 25-column analytical table:

- **Rows:** 45,593 → 45,502 (removed ≈91 unsalvageable/duplicate records).
- **Type coercion:** I coerced `Delivery_person_Age` and `Delivery_person_Ratings` from string-encoded "NaN" sentinels into proper numeric values.

- **Feature engineering:** I engineered `distance_km` (Haversine distance derived from the four lat/long columns), `day`, `month`, `day_of_week`, `is_weekend` (parsed from `Order_Date`), `pickup_time` & `order_time_hour`, and `time_of_day` — plus `city_type`/`city_name` extracted from the ID fields.

3. Data Quality at a Glance

Field	% Missing	Note
`age`	4.1%	coerced from string
`ratings`	4.2%	coerced from string
`restaurant_lat/long`, `delivery_lat/long`, `distance_km`	8.0%	missing together
`weather`	1.0%	
`traffic`	1.0%	
`multiple_deliveries`	2.2%	
`festival`	0.5%	
`city_type`	2.6%	
`pickup_time` / `order_time_hour`	3.6%	
`time_of_day`	4.5%	
`time` (target)	0.0%	fully observed

4. Univariate Findings

Numerical variables

Variable	Mean	Median	Std	Skew	Kurtosis	Observation
`age`	29.6	30	5.76	−0.01	−1.21	Flat/uniform across 20–39; platykurtic (no peak)
`ratings`	4.64	4.7	0.31	−1.79	5.14	Heavily left-skewed; small but real cluster of low-rated riders (≤4.0)
`time` (target)	26.3	26	9.39	0.49	−0.31	Bimodal (peaks ~17–18 min and ~26–27 min), long right tail to 54 min
`distance_km`	9.72	9.19	5.60	0.32	−0.91	Bimodal with a **gap between 13–16 km**

Categorical variables (share of orders)

Variable	Distribution
`traffic`	low 34%, jam 31%, medium 24%, high 10%
`weather`	~equal across 6 conditions (16–17% each)
`vehicle_type`	motorcycle 58%, scooter 34%, e-scooter 8%, **bicycle 0.1%**
`vehicle_condition`	conditions 1/2/3 ≈ 33% each, **condition 0 just 0.9%**
`multiple_deliveries`	1 order 63%, 0 orders (solo trip, no batching) 32%, 2 orders 4.5%, 3 orders 0.8%
`festival`	no 98%, **yes 2%**
`city_type`	metropolitan 77%, urban 23%, **semi-urban 0.4%**
`order_type`	snack/meal/drinks/buffet ≈ 25% each
`is_weekend`	weekday 75%, weekend 25%
`time_of_day`	evening 42%, night 21%, morning 20%, afternoon 17%
`day` of month	**day 3 (28%) and day 4 (14%) alone = 42%** of all orders
`month`	**Feb + March = ~63%** of all orders

I identified that the `day/month` concentration is highly artificial. Approximately 42% of all orders fall on just two calendar days, indicating this is an artifact of the data extraction window rather than genuine business seasonality.

5. What Drives Delivery Time? (Bivariate)

I tested all categorical-vs-time comparisons using ANOVA (3+ groups) or a t-test (2 groups). Statistical significance ($p < 0.05$) was found for every variable **except `order_type` (p = 0.225)** and `pickup_time` ($p = 0.188$). Both can reasonably be excluded as meaningful operational drivers.

- **A. Operational / environmental conditions (The biggest levers):** Traffic acts as the strongest single categorical driver, climbing from ~20 min (low) to ~30 min (jam). Weather conditions show critical differences in the *tails* (sunny weather exhibits extreme high-traffic outliers; stormy/sandstorm/windy conditions behave similarly as "adverse weather").
- **B. Fleet & rider factors:** Electric scooters are definitively the fastest, while motorcycles are the slowest and exhibit the most variance.
- **C. Demand-side factors:** Delivery time rises steadily as batch size increases (1 → 2 → 3 stops), landing in the 35–50 minute range for highly batched orders. Festival days jump to 45 mins.
- **D. Temporal factors:** The evening is simultaneously the highest-volume (42% of orders) and the slowest (median ~28–29 min) window, representing a textbook supply/demand mismatch. Weekend orders are marginally faster overall compared to weekdays.

6. Business Actions

Based purely on the knowledge gained from this EDA, the following operational changes are required:

- **Implement Dynamic SLA / ETA communication:** The static ETA promise must be replaced with a traffic-and-weather-aware ETA range (e.g., low traffic ≈ 20 min, jam ≈ 30 min, jam + adverse weather ≈ 35–40 min).
- **Execute the Festival Staffing Playbook:** Because festival days are predictable and reliably ~80% slower, the operations team will pre-position extra riders and notify customers proactively.
- **Revise Batching Policies:** Multiple-delivery orders trade per-order speed for overall fleet efficiency. I have determined that batching 3 orders causes severe SLA breaches, and the dispatch algorithm must limit batching to a maximum of 2 orders.
- **Adjust the Fleet Mix:** Electric scooters exhibit the most favorable delivery time profile. Fleet expansion will prioritize electric scooters in applicable zones.
- **Fix Data Collection:** Engineering must close the missing GPS coordinate gap (8% of orders) immediately to ensure reliable distance calculations in our backend.